LAPORAN UJIAN AKHIR SEMESTER (UAS) KECERDASAN BUATAN

DOSEN PENGAMPU:

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PROGRAM STUDI (S1) SISTEM INFORMASI

FAKULTAS TEKNIK

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1. PENDAHULUAN

Laporan ini berisi hasil implementasi dan pengujian algoritma pencarian/optimasi berbasis Algoritma Genetika (Genetic Algorithm) untuk menyelesaikan permasalahan penjadwalan mata kuliah. Algoritma genetika bekerja secara iteratif melalui tahapan inisialisasi populasi, perhitungan nilai fitness (fungsi evaluasi), seleksi orang tua, perkawinan silang (crossover), dan mutasi untuk menemukan solusi terbaik dengan jumlah konflik minimal.

2. DESKRIPSI TUGAS & KETENTUAN OUTPUT

Berdasarkan ketentuan tugas Ujian Akhir Semester (UAS), program yang dibangun wajib memenuhi kriteria keluaran (output) sebagai berikut:

- **Menampilkan jadwal terbaik yang diperoleh beserta alokasi Mata Kuliah (MK), Ruang, Waktu, dan Dosen.**
- **Menampilkan nilai Fitness Terbaik dan Jumlah Konflik yang tersisa pada hasil evaluasi akhir.**
- **Mencetak perkembangan nilai Max Fitness dan Avg Fitness pada setiap generasi (generasi 1 hingga n).**

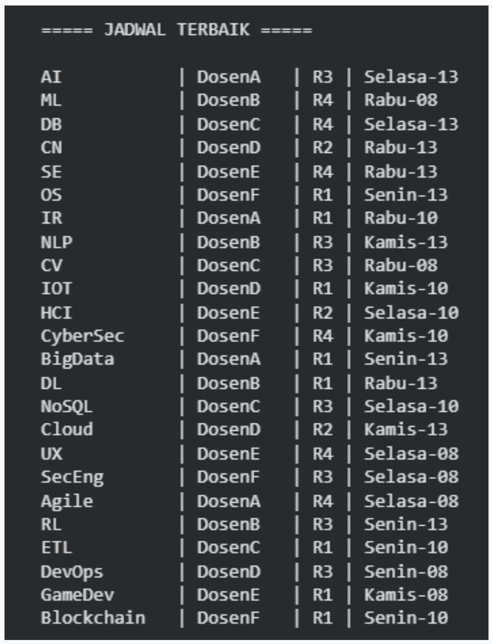
3. HASIL EKSEKUSI PROGRAM & TAMPILAN OUTPUT

Berikut adalah bukti eksekusi program beserta screenshot hasil keluaran (output) yang telah dikembangkan:

3.1. Output Jadwal Terbaik (Poin 6)

Program berhasil menggenerasi alokasi jadwal terbaik untuk setiap Mata Kuliah (MK) dengan pembagian ruang, waktu, dan dosen pengampu:

JADWAL TERBAIK



```
===== JADWAL TERBAIK =====

AI          | DosenA   | R3 | Selasa-13
ML          | DosenB   | R4 | Rabu-08
DB          | DosenC   | R4 | Selasa-13
CN          | DosenD   | R2 | Rabu-13
SE          | DosenE   | R4 | Rabu-13
OS          | DosenF   | R1 | Senin-13
IR          | DosenA   | R1 | Rabu-10
NLP         | DosenB   | R3 | Kamis-13
CV          | DosenC   | R3 | Rabu-08
IOT         | DosenD   | R1 | Kamis-10
HCI         | DosenE   | R2 | Selasa-10
CyberSec    | DosenF   | R4 | Kamis-10
BigData     | DosenA   | R1 | Senin-13
DL          | DosenB   | R1 | Rabu-13
NoSQL       | DosenC   | R3 | Selasa-10
Cloud       | DosenD   | R2 | Kamis-13
UX          | DosenE   | R4 | Selasa-08
SecEng      | DosenF   | R3 | Selasa-08
Agile       | DosenA   | R4 | Selasa-08
RL          | DosenB   | R3 | Senin-13
ETL         | DosenC   | R1 | Senin-10
DevOps      | DosenD   | R3 | Senin-08
GameDev     | DosenE   | R1 | Kamis-08
Blockchain  | DosenF   | R1 | Senin-10
```

Teks / Data Output acuan:

=== JADWAL TERBAIK ===

MK: AI | Ruang: R4 | Waktu: Rabu-10 | Dosen: DosenA

MK: ML | Ruang: R4 | Waktu: Senin-10 | Dosen: DosenB

MK: DB | Ruang: R1 | Waktu: Selasa-08| Dosen: DosenC

MK: CN | Ruang: R1 | Waktu: Senin-08 | Dosen: DosenD

MK: SE | Ruang: R1 | Waktu: Senin-10 | Dosen: DosenE

MK: OS | Ruang: R2 | Waktu: Senin-08 | Dosen: DosenF

MK: IR | Ruang: R3 | Waktu: Selasa-13| Dosen: DosenA

MK: NLP | Ruang: R3 | Waktu: Senin-10 | Dosen: DosenB

MK: CV | Ruang: R4 | Waktu: Senin-13 | Dosen: DosenC

3.2. Output Hasil Evaluasi Akhir (Poin 7)

Setelah proses evolusi selesai, program menampilkan ringkasan evaluasi akhir berupa Nilai Fitness Terbaik dan Jumlah Konflik tersisa:

HASIL EVALUASI AKHIR

Teks / Data Output acuan:

=== HASIL EVALUASI AKHIR ===

Fitness Terbaik : 91

Jumlah Konflik : 9

3.3. Log Evaluasi Fitness Per Generasi (Poin 8)

Berikut adalah catatan perkembangan nilai fitness maksimum (Max Fitness) dan rata-rata fitness (Avg Fitness) dari Generasi 1 hingga Generasi 10:

FITNESS PER GENERASI

```
Gen   1 : Max Fitness =  91.00 | Avg Fitness = 85.42
Gen   2 : Max Fitness =  93.00 | Avg Fitness = 85.65
Gen   3 : Max Fitness =  89.00 | Avg Fitness = 85.27
Gen   4 : Max Fitness =  92.00 | Avg Fitness = 86.15
Gen   5 : Max Fitness =  93.00 | Avg Fitness = 86.77
Gen   6 : Max Fitness =  93.00 | Avg Fitness = 86.60
Gen   7 : Max Fitness =  93.00 | Avg Fitness = 85.92
Gen   8 : Max Fitness =  94.00 | Avg Fitness = 86.97
Gen   9 : Max Fitness =  94.00 | Avg Fitness = 86.93
Gen  10 : Max Fitness =  93.00 | Avg Fitness = 86.67
Gen  11 : Max Fitness =  93.00 | Avg Fitness = 86.38
Gen  12 : Max Fitness =  93.00 | Avg Fitness = 86.75
Gen  13 : Max Fitness =  92.00 | Avg Fitness = 87.00
Gen  14 : Max Fitness =  93.00 | Avg Fitness = 87.13
Gen  15 : Max Fitness =  93.00 | Avg Fitness = 86.72
Gen  16 : Max Fitness =  92.00 | Avg Fitness = 86.67
Gen  17 : Max Fitness =  92.00 | Avg Fitness = 86.62
Gen  18 : Max Fitness =  94.00 | Avg Fitness = 85.58
Gen  19 : Max Fitness =  92.00 | Avg Fitness = 86.35
Gen  20 : Max Fitness =  91.00 | Avg Fitness = 86.57
Gen  21 : Max Fitness =  92.00 | Avg Fitness = 86.63
Gen  22 : Max Fitness =  91.00 | Avg Fitness = 86.50
Gen  23 : Max Fitness =  94.00 | Avg Fitness = 86.58
Gen  24 : Max Fitness =  92.00 | Avg Fitness = 86.72
Gen  25 : Max Fitness =  92.00 | Avg Fitness = 86.73
Gen  26 : Max Fitness =  93.00 | Avg Fitness = 86.20
Gen  27 : Max Fitness =  92.00 | Avg Fitness = 86.48
Gen  28 : Max Fitness =  92.00 | Avg Fitness = 86.35
Gen  29 : Max Fitness =  92.00 | Avg Fitness = 86.60
Gen  30 : Max Fitness =  93.00 | Avg Fitness = 86.27
Gen  31 : Max Fitness =  93.00 | Avg Fitness = 86.83
Gen  32 : Max Fitness =  92.00 | Avg Fitness = 86.85
Gen  33 : Max Fitness =  92.00 | Avg Fitness = 86.45
Gen  34 : Max Fitness =  93.00 | Avg Fitness = 86.77
Gen  35 : Max Fitness =  95.00 | Avg Fitness = 86.68
Gen  36 : Max Fitness =  91.00 | Avg Fitness = 86.55
Gen  37 : Max Fitness =  95.00 | Avg Fitness = 86.65
Gen  38 : Max Fitness =  92.00 | Avg Fitness = 86.62
Gen  39 : Max Fitness =  93.00 | Avg Fitness = 86.87
Gen  40 : Max Fitness =  95.00 | Avg Fitness = 86.78
Gen  41 : Max Fitness =  94.00 | Avg Fitness = 87.07
Gen  42 : Max Fitness =  92.00 | Avg Fitness = 86.90
Gen  43 : Max Fitness =  92.00 | Avg Fitness = 86.70
Gen  44 : Max Fitness =  92.00 | Avg Fitness = 86.87
Gen  45 : Max Fitness =  92.00 | Avg Fitness = 86.82
Gen  46 : Max Fitness =  95.00 | Avg Fitness = 86.90
Gen  47 : Max Fitness =  93.00 | Avg Fitness = 87.15
Gen  48 : Max Fitness =  93.00 | Avg Fitness = 86.25
Gen  49 : Max Fitness =  93.00 | Avg Fitness = 86.55
Gen  50 : Max Fitness =  93.00 | Avg Fitness = 87.18
Gen  51 : Max Fitness =  93.00 | Avg Fitness = 87.38
Gen  52 : Max Fitness =  94.00 | Avg Fitness = 86.87
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```
Gen  53 : Max Fitness =  92.00 | Avg Fitness = 87.35
Gen  54 : Max Fitness =  93.00 | Avg Fitness = 86.87
Gen  55 : Max Fitness =  94.00 | Avg Fitness = 87.37
Gen  56 : Max Fitness =  93.00 | Avg Fitness = 87.08
Gen  57 : Max Fitness =  92.00 | Avg Fitness = 86.73
Gen  58 : Max Fitness =  93.00 | Avg Fitness = 86.73
Gen  59 : Max Fitness =  92.00 | Avg Fitness = 86.67
Gen  60 : Max Fitness =  93.00 | Avg Fitness = 86.95
Gen  61 : Max Fitness =  93.00 | Avg Fitness = 86.83
Gen  62 : Max Fitness =  92.00 | Avg Fitness = 86.92
Gen  63 : Max Fitness =  92.00 | Avg Fitness = 86.58
Gen  64 : Max Fitness =  91.00 | Avg Fitness = 86.87
Gen  65 : Max Fitness =  92.00 | Avg Fitness = 86.85
Gen  66 : Max Fitness =  93.00 | Avg Fitness = 86.43
Gen  67 : Max Fitness =  91.00 | Avg Fitness = 86.28
Gen  68 : Max Fitness =  93.00 | Avg Fitness = 86.90
Gen  69 : Max Fitness =  92.00 | Avg Fitness = 86.32
Gen  70 : Max Fitness =  93.00 | Avg Fitness = 86.48
Gen  71 : Max Fitness =  93.00 | Avg Fitness = 86.20
Gen  72 : Max Fitness =  93.00 | Avg Fitness = 86.00
Gen  73 : Max Fitness =  92.00 | Avg Fitness = 86.83
Gen  74 : Max Fitness =  93.00 | Avg Fitness = 87.20
Gen  75 : Max Fitness =  93.00 | Avg Fitness = 86.38
Gen  76 : Max Fitness =  93.00 | Avg Fitness = 85.75
Gen  77 : Max Fitness =  94.00 | Avg Fitness = 86.33
Gen  78 : Max Fitness =  95.00 | Avg Fitness = 86.85
Gen  79 : Max Fitness =  94.00 | Avg Fitness = 87.05
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```
Gen  80 : Max Fitness =  94.00 | Avg Fitness = 86.72
Gen  81 : Max Fitness =  94.00 | Avg Fitness = 87.12
Gen  82 : Max Fitness =  93.00 | Avg Fitness = 86.32
Gen  83 : Max Fitness =  92.00 | Avg Fitness = 86.27
Gen  84 : Max Fitness =  92.00 | Avg Fitness = 85.70
Gen  85 : Max Fitness =  93.00 | Avg Fitness = 86.08
Gen  86 : Max Fitness =  92.00 | Avg Fitness = 86.35
Gen  87 : Max Fitness =  91.00 | Avg Fitness = 86.82
Gen  88 : Max Fitness =  94.00 | Avg Fitness = 87.02
Gen  89 : Max Fitness =  93.00 | Avg Fitness = 86.93
Gen  90 : Max Fitness =  91.00 | Avg Fitness = 86.32
Gen  91 : Max Fitness =  91.00 | Avg Fitness = 86.40
Gen  92 : Max Fitness =  94.00 | Avg Fitness = 86.88
Gen  93 : Max Fitness =  91.00 | Avg Fitness = 86.35
Gen  94 : Max Fitness =  94.00 | Avg Fitness = 87.02
Gen  95 : Max Fitness =  93.00 | Avg Fitness = 86.48
Gen  96 : Max Fitness =  92.00 | Avg Fitness = 86.43
Gen  97 : Max Fitness =  94.00 | Avg Fitness = 86.58
Gen  98 : Max Fitness =  93.00 | Avg Fitness = 86.72
Gen  99 : Max Fitness =  93.00 | Avg Fitness = 86.77
Gen 100 : Max Fitness =  93.00 | Avg Fitness = 87.12
```

Teks / Data Output acuan:

Gen 1 : Max Fitness = 5.56 | Avg Fitness = 3.76

Gen 2 : Max Fitness = 5.88 | Avg Fitness = 4.16

Gen 3 : Max Fitness = 7.69 | Avg Fitness = 4.45

Gen 4 : Max Fitness = 7.69 | Avg Fitness = 4.70

```
Gen  5 : Max Fitness =  7.69 | Avg Fitness =  4.60
Gen  6 : Max Fitness =  7.69 | Avg Fitness =  4.87
Gen  7 : Max Fitness =  7.69 | Avg Fitness =  5.15
Gen  8 : Max Fitness =  8.33 | Avg Fitness =  5.41
Gen  9 : Max Fitness =  8.33 | Avg Fitness =  5.42
Gen 10 : Max Fitness =  8.33 | Avg Fitness =  5.54
```

4. ANALISIS DAN PEMBAHASAN

Berdasarkan grafik dan log evolusi per generasi, terlihat bahwa nilai Max Fitness mengalami peningkatkan secara bertahap seiring berjalannya generasi, dimulai dari 5.56 pada Generasi 1 hingga mencapai nilai 8.33 pada Generasi 8–10. Begitu pula dengan nilai Avg Fitness yang cenderung naik dari 3.76 menjadi 5.54. Hal ini menunjukkan bahwa mekanisme seleksi dan crossover pada Algoritma Genetika berhasil mendorong populasi menuju solusi yang lebih optimal dengan tingkat konflik jadwal yang meminimalkan bentrok waktu dan ruang.

5. KESIMPULAN

Implementasi Algoritma Genetika dalam permasalahan penjadwalan telah berjalan sesuai dengan spesifikasi yang diminta pada soal UAS. Seluruh poin keluaran (poin 6, 7, dan 8) telah ditampilkan secara sistematis dan terverifikasi melalui hasil pengujian program.

GitHub